

The Feature Tokenised Transformer with Cross Sectional Coupling

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Abstract

This document describes a feature tokenised transformer with cross sectional coupling for monthly portfolio construction, following the feature tokenised transformer of Gorishniy et al. (2021). Each security’s characteristics are tokenised per feature and passed through transformer blocks that attend across the feature tokens, after which the pooled per security representation is coupled across the cross section through an explicit stock to market and market to stock exchange. The network is fitted under a maximum Sharpe ratio regression objective. We develop the full model, present the tokenisation, the attention over features, and the coupling, and interpret the realised performance.

1 Overview and motivation

A model on this panel may exploit two kinds of interaction. The first is the interaction among the characteristics of a single security, which determines how the individual signals combine into an assessment of that security. The second is the interaction across securities, which places that assessment in the context of the month. A plain perceptron treats the first with a feedforward encoder and the second with a pooled context. The present model enriches the first interaction by tokenising each feature and attending across the tokens, while retaining an explicit cross sectional coupling for the second. The purpose is to test whether a more expressive per security encoder, of a kind that performs well on tabular problems, improves the portfolio.

2 The data pipeline and objective

The raw inputs are a panel of monthly characteristics and a list of candidate features, and the target is the excess return of a security over the following month. A characteristic is retained if its coverage is at least ninety percent over the pre test period, which makes the selection causal and algorithmic. For each month every retained characteristic is transformed to its within month percentile rank, missing entries are set to the median before centring so that the inputs lie in the interval $[-0.5, 0.5]$, and securities missing more than one third of the characteristics are dropped. The network is fitted under the loss $(1 - \sum_i w_i r_i)^2$, which drives the realised managed return towards unity in every month, on a rolling window of at most sixty months, with a fifty epoch cold start, ten epoch warm starts thereafter, three seeds, and seed averaging of the weights normalised to unit gross exposure.

3 Feature tokenisation and attention over features

Each of a security’s characteristics is turned into a token by scaling a learned embedding by the feature value and adding a learned bias. Writing x_{ij} for the value of security i on characteristic j ,

the token is

$$t_{ij} = x_{ij} e_j + b_j, \quad e_j, b_j \in \mathbb{R}^{d_t}, \quad (1)$$

and a learned classification token t_{i0} is prepended to the sequence of feature tokens. The sequence of one security is passed through a small number of transformer blocks, each of which applies multihead self attention across the feature tokens followed by a position wise feedforward network, both with residual connections and layer normalisation,

$$t \leftarrow t + \text{attn}(\text{LN}(t)), \quad t \leftarrow t + \text{ff}(\text{LN}(t)). \quad (2)$$

The attention here is over the feature tokens of a single security, so it learns interactions among that security's characteristics. The pooled per security representation is taken as the classification token after the blocks, $z_i = \text{LN}(t_{i0})$.

4 Cross sectional coupling on the pooled representation

The second interaction acts across securities. A stock to market map pools the representations,

$$m = \frac{1}{n} \sum_{i=1}^n \text{to_market}(z_i), \quad (3)$$

and a market to stock map broadcasts the market state back to each security,

$$z_i \leftarrow z_i + \text{from_market}([z_i, m]), \quad (4)$$

after which the score is read from a head on z_i . In this way the attention is spent on the interactions among a security's characteristics, which is where a transformer is expected to help, while the cross sectional coupling is handled by the explicit market exchange, which lets a security's score depend on the composition of its month.

5 Evaluation and concluding remarks

On the evaluation window the model attains a Sharpe ratio of 2.050 in the volatility scaled form and 2.161 in the unscaled form. A plainer network with a feedforward encoder and a pooled coupling attains 2.484 on this panel, so the more expressive per security encoder does not improve the portfolio here, even though the cross sectional coupling is preserved. We attribute this to the cause that limits other high capacity encoders on this panel, namely that the additional parameters of the transformer are estimated from a limited window and add variance to the estimated map without a commensurate gain in the signal. We conclude that on this panel the return relevant content of the characteristics is captured adequately by a plain encoder, and that the value of the construction lies in the cross sectional coupling rather than in the sophistication of the per security representation.